



University of Sri Jayewardenepura
Faculty of Applied Sciences
Department of Computer Science

B.Sc. General Degree First Year
First Semester Terminal Course Unit Examination – September 2025
Information and Communication Technology

ICT 1032 Software Architecture and Design

Instructions

This paper contains **4 questions**. Answer **all questions**. Each question carries 25 marks.

Time: Two (2) hours

Question 1

Part A

- i) Differentiate between Software Architecture and Software Design in terms of *scope*.
- ii) List and briefly explain two types of quality attributes in software systems.
- iii) Explain how an architectural style provides a framework for system design.

[3 x 3 marks]

Part B State whether the following statements are true or false. *For each case briefly describe your answer.*

- i) Architectural style focuses on the detailed algorithms and workflows of a system.
- ii) Software architecture always includes private implementation details of components.
- iii) Software requirements alone are always sufficient to design an architecture.

[3 x 2 marks]

Part C Consider the following scenario to answer the questions in parts i) and ii).

A hospital is planning to develop a patient management system. The initial requirements include storing patient details, managing appointments, and generating reports. However, the system must also support future expansion, handle sensitive medical data securely, and work reliably under heavy usage.

- i) Identify and explain two architectural decisions (business or technical) that would strongly influence the quality attributes of this system.

- ii) Suggest one suitable architectural style for this system and justify your choice.

[2 x 5 marks]

Question 2

Part A

- i) Briefly explain three main reasons why software architecture is important in software development.
- ii) List the three main groups of architectural structures and provide one example for each.
- iii) Briefly describe the Layered Architecture style and explain three benefits of organizing a system into layers.

[3 x 3 marks]

Part B State whether the following statements are true or false. *For each case briefly describe your answer.*

- i) A reference model and a reference architecture are identical.
- ii) Using an architectural pattern like client-server automatically guarantees high system performance.
- iii) Software architecture is just the source code of a system written in a structured way.

[3 x 2 marks]

Part C Consider the following scenario to answer the questions in parts i) and ii).

A software company is developing a **smart city traffic management system**. The system must,

- Process real-time data from thousands of sensors and cameras.
- Allow modular development by different teams (data collection, analytics, reporting).
- Run on distributed servers for scalability and reliability.
- Be easily maintainable and support future upgrades.

- i) Identify and briefly explain **two architectural structures** that would be most useful for this system.
- ii) Explain how the **Architecture Business Cycle (ABC)** could influence the design of this system, mentioning with **two external factors**.

[2 x 5 marks]

Question 3

Part A

- i) Briefly explain the three categories of Data-Flow Architectures with one example for each.
- ii) Differentiate between a Logical Data Flow Diagram and a Physical Data Flow Diagram.

- iii) List and briefly describe the four main components of a Data Flow Diagram (DFD).

[3 x 3 marks]

Part B State whether the following statements are true or false. *For each case briefly describe your answer.*

- i) In a Batch Sequential System, subsystems can start processing before the previous one finishes.
- ii) In a Data Flow Diagram, every process must have at least one input and one output data flow.
- iii) In a Data-Flow Architecture, the flow of control determines the execution order of components.

[3 x 2 marks]

Part C Consider the following scenario to answer the question in parts i).

A library management system is being designed to automate book borrowing and returning. The system must,

- Allow students to search for books and borrow them.
- Update the book inventory when a book is borrowed or returned.
- Maintain student records, including borrowed books and due dates.
- Generate fine details for overdue books.

- i) Draw a **Context Diagram** and a **Level-0 Data Flow Diagram (DFD)** for the library management system.

[10 marks]

Question 4

Part A

- i) Briefly explain the role of the Enterprise Service Bus (ESB) in Service-Oriented Architecture (SOA).
- ii) List any three challenges of Microservices Architecture.
- iii) State three best practices for writing effective software documentation.

[3 x 3 marks]

Part B State whether the following statements are true or false. *For each case briefly describe your answer.*

- i) In SOA, services are tightly coupled, which makes them harder to reuse.
- ii) Microservices architecture always guarantees simpler debugging compared to monolithic systems.

- iii) Cloud services based on SOA or Microservices allow independent scaling of components to handle variable workloads.

[3 x 2 marks]

Part C Consider the following scenario to answer the questions in parts i) and ii).

A healthcare provider is building a new patient portal system. The system must,

- Allow patients to access lab results and appointment schedules.
- Integrate securely with existing hospital databases and external pharmacy services.
- Scale to support thousands of concurrent users.

i) Recommend whether SOA or Microservices would be a better fit for this system. Justify your answer.

ii) Identify and explain two key quality attributes that this architecture would improve.

[2 x 5 marks]

END OF PAPER